

Trainings ▾

Preparatory Steps

Automotive with CM875, CM870 and CM570

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children or pets. If not reused, dispose of in accordance with local environmental regulations.

- Drain the cooling system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html>).
- Remove the air intake piping. Refer to the OEM troubleshooting and repair manual.
- Remove the air compressor air supply tube from the EGR mixer and the air compressor air supply tube at the rear of the engine. Refer to Procedure 012-109 (</qs3/pubsys2/xml/en/procedures/35/35-012-109-tr.html>).
- Remove the exhaust gas pressure sensor tube. Refer to Procedure 011-027 (</qs3/pubsys2/xml/en/procedures/35/35-011-027-tr.html>).
- Remove the EGR valve coolant supply tube from the EGR cooler upper supply tube. Refer to Procedure 011-030 (</qs3/pubsys2/xml/en/procedures/35/35-011-030-tr.html>).
- Remove the turbocharger coolant return tube from the turbocharger. Disconnect the turbocharger coolant supply tube from the EGR cooler coolant supply tube. Refer to Procedure 010-041

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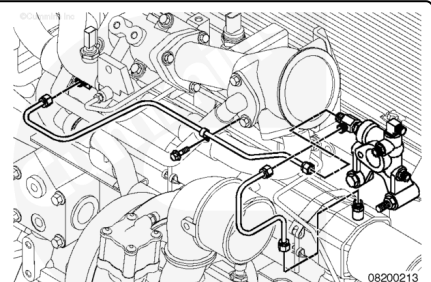
- Drain the cooling system. Refer to Procedure 008-018 (/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html).
- Remove the air intake piping. Refer to the OEM troubleshooting and repair manual.
- Remove the turbocharger coolant supply tube from the turbocharger. Refer to Procedure 010-041 (/qs3/pubsys2/xml/en/procedures/35/35-010-041-tr.html).

Remove

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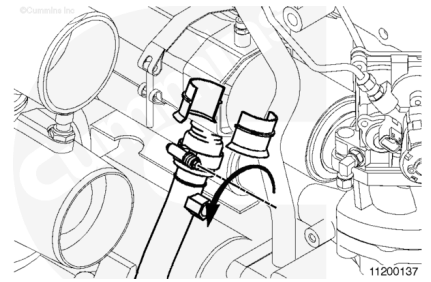
Remove the EGR cooler coolant rear vent tube retaining clip.

Remove the EGR cooler coolant front and rear vent tubes.



Loosen the hose clamp which secures the tube retainers to the EGR cooler and EGR cooler coolant supply upper tube.

Remove the tube retainers.



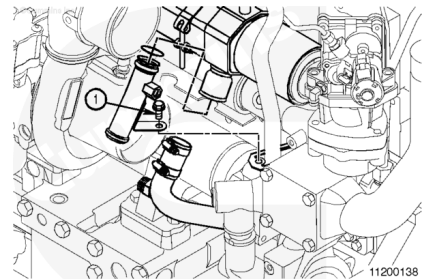
Remove the capscrew (1), which secures the EGR cooler coolant supply upper tube as well as the water manifold return tube to the thermostat housing support.

Removal of the water manifold return tube is **not** required, as the upper tube and brace can be rotated from underneath the water manifold return tube.

Loosen the two hose clamps on the EGR cooler coolant supply hose.

Remove the EGR cooler coolant supply upper tube from the EGR cooler coolant supply lower tube and the EGR cooler coolant supply hose.

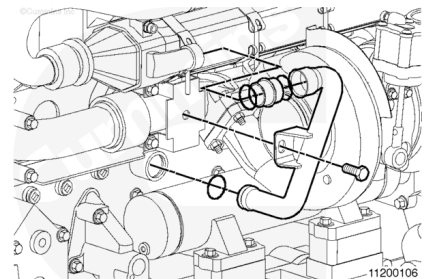
Remove and discard the o-rings from the EGR cooler coolant supply upper tube.



Remove the EGR cooler coolant return tube mounting capscrew.

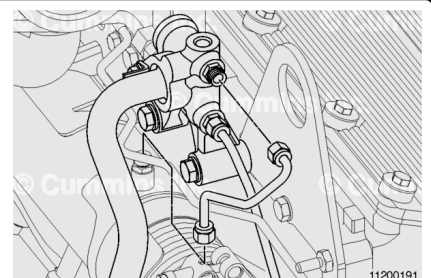
Disconnect the EGR cooler coolant return jumper connection and EGR cooler coolant return tube assembly from the EGR cooler while removing the coolant return tube from the water header plate. Remove and discard the o-rings.

Remove the EGR cooler coolant return jumper connection. Remove and discard the o-rings.



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Remove the EGR cooler coolant vent tube.

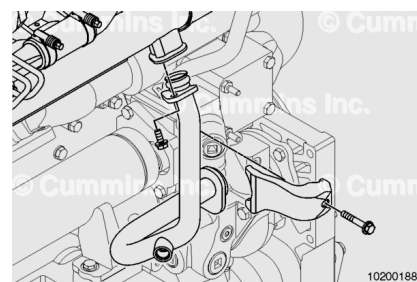


Remove the EGR cooler coolant supply tube mounting capscrew from the EGR cooler.

Remove the EGR cooler coolant supply tube mounting clamp and capscrew from the water pump.

Remove the EGR cooler coolant supply tube.

Remove and discard the o-rings from the EGR cooler coolant supply tube.

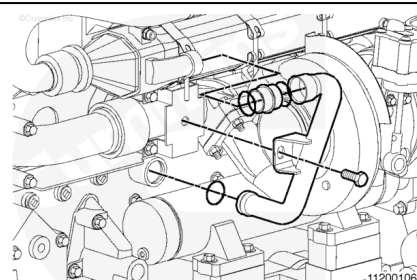


Remove the EGR cooler coolant return tube mounting capscrews.

Disconnect the EGR cooler coolant return jumper connection and EGR cooler coolant return tube assembly from the EGR cooler while removing the coolant return tube from the water header plate. Remove and discard the o-rings.

Remove the EGR cooler coolant return jumper connection.

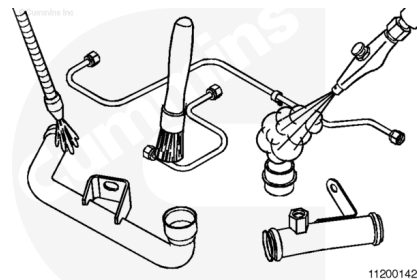
Remove and discard the o-rings.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvent, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

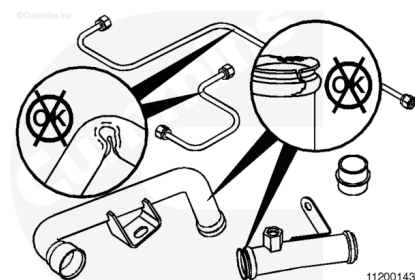
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the EGR cooler coolant vent tubes, EGR cooler coolant supply tube, EGR coolant return tube, and jumper connection with solvent.

Dry with compressed air.

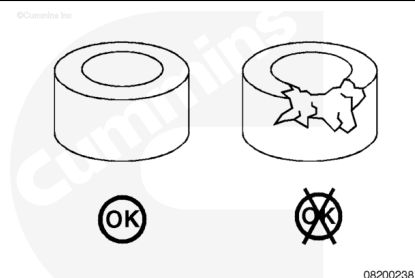
Inspect the EGR cooler coolant vent tubes, EGR coolant supply tube, EGR coolant return tube, and jumper connection for cracks or any other damage.

Replace the component if cracked or otherwise damaged.



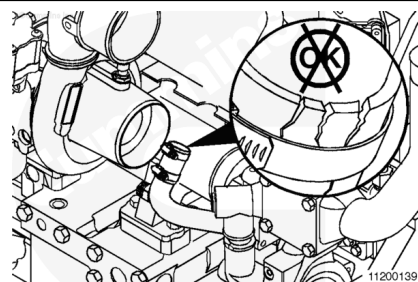
Inspect the grommets for cracks or other damage.

Replace the grommets if they are cracked or otherwise damaged.



Note : The following check does not apply to automotive with CM876.

Check the EGR cooler coolant supply hose for cracks and tears. Replace if damaged.



Install

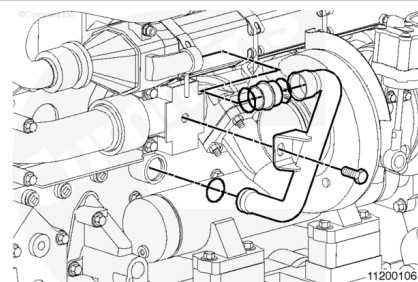
Automotive with CM875, CM870 and CM570

⚠ CAUTION ⚠

Do not use petroleum products on the coolant plumbing o-rings. The o-rings will swell and cause the connection to leak if petroleum products are used.

Lubricate the o-ring connection mating surfaces with clean engine coolant, soapy water, or vegetable oil. To avoid pinching or rolling the o-rings during assembly, do **not** lubricate the o-ring.

Install new o-rings on the EGR cooler coolant return tube and jumper connection.



Install the EGR cooler coolant return jumper connection into the EGR cooler coolant return tube.

Position the lower end of the EGR cooler coolant return tube jumper connection as illustrated in the graphic.

Install the EGR cooler coolant return tube from the water header plate to the EGR cooler coolant jumper connection.

Coat the EGR cooler coolant return tube mounting capscrew with high temperature anti-seize compound.

Install and tighten the capscrew.

Torque Value: 47 n·m [35 ft-lb]

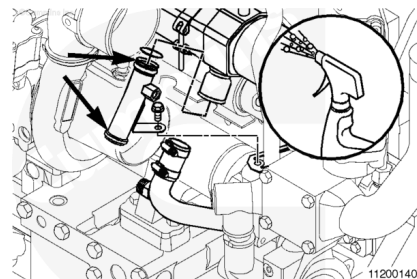
Lubricate the o-ring connection mating surfaces with clean engine coolant, soapy water, or vegetable oil. To avoid pinching or rolling the o-rings during assembly, do **not** lubricate the o-ring.

Install new o-rings on the EGR cooler coolant supply upper tube.

Do **not** lubricate the retainer o-ring. Install the retainer o-ring (dry) onto the EGR cooler coolant supply upper tube past the small o-ring gland.

Install the EGR cooler coolant supply upper tube from the EGR cooler to the EGR cooler coolant supply lower tube and the EGR cooler coolant supply hose.

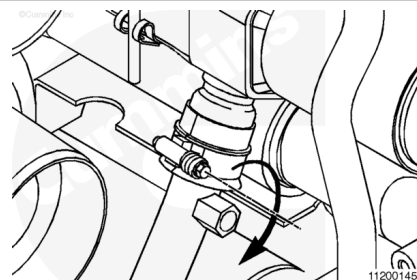
Install the EGR cooler coolant supply upper tube into the EGR cooler flange. Push the retainer o-ring up the EGR cooler coolant supply tube until it contacts the EGR cooler flange.



Install the tube retainers and hose clamp onto the EGR cooler coolant supply upper tube, EGR cooler and retainer o-ring.

Tighten the hose clamp.

Torque Value: 5 n·m [44 in-lb]

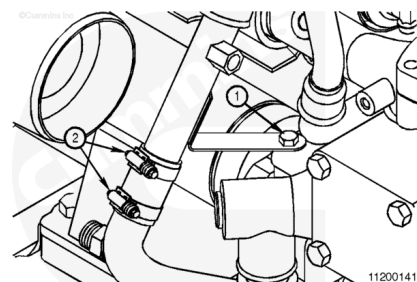


Tighten the EGR cooler coolant supply upper tube and water manifold return tube mounting capscrew (1).

Torque Value: 20 n·m [177 in-lb]

Equally space the EGR cooler coolant supply hose over the EGR cooler coolant supply upper and lower tubes and tighten the hose clamps (2).

Torque Value: 5 n•m [44 in-lb]

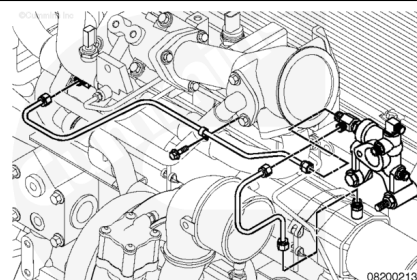


Lubricate the grommets with clean engine coolant, soapy water, or vegetable oil for easier installation and to avoid pinching or rolling the grommets while assembling.

Install the EGR cooler coolant front and rear vent tubes.

Install the retaining clip and capscrew for the EGR cooler coolant rear vent tube and tighten the capscrew.

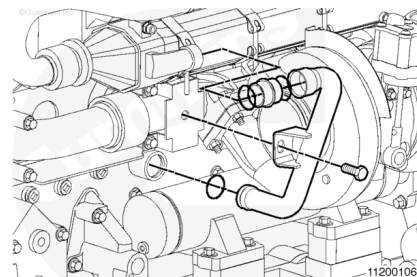
Torque Value: 47 n•m [35 ft-lb]



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Do not use petroleum products on the coolant plumbing o-rings. The o-rings will swell and cause the connection to leak if petroleum products are used.



Lubricate the o-ring connection mating surfaces with clean engine coolant, soapy water, or vegetable oil. To avoid pinching or rolling the o-rings during assembly, do **not** lubricate the o-ring.

Install new o-rings on the EGR cooler coolant return tube and jumper connection.

Install the EGR cooler coolant return jumper connection into the EGR cooler coolant return tube.

Position the lower end of the EGR cooler coolant return tube jumper connection as illustrated in the graphic.

Install the EGR cooler coolant return tube from the water header plate to the EGR cooler coolant jumper connection.

Coat the EGR cooler coolant return tube mounting capscrew with high temperature anti-seize compound.

Install and tighten the capscrew.

Torque Value: 47 n•m [35 ft-lb]

Lubricate the o-ring connection mating surfaces with clean engine coolant, soapy water, or vegetable oil. To avoid pinching or rolling the o-rings during assembly, do **not** lubricate the o-ring.

Install new o-rings on the EGR cooler coolant supply tube.

Install the EGR cooler coolant supply tube into the EGR cooler and the water pump.

Install the EGR cooler coolant supply tube capscrew into the EGR cooler coolant inlet flange. Do **not** torque.

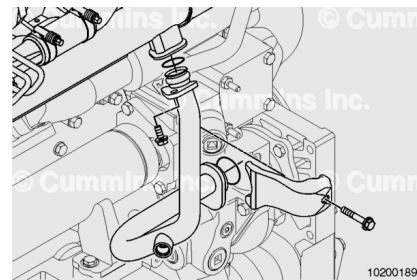
Make sure the EGR cooler coolant supply tube is seated completely in the water pump body before tightening the EGR cooler coolant supply tube retaining clamp capscrew to reduce the possibility of pinching the o-ring.

Tighten the remaining clamp capscrew.

Torque Value: 34 n•m [25 ft-lb]

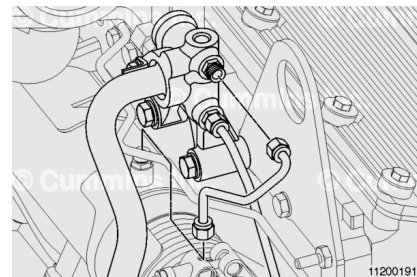
Tighten the EGR cooler coolant supply tube upper mounting capscrew.

Torque Value: 34 n•m [25 ft-lb]



Lubricate the grommets with clean engine coolant, soapy water, or vegetable oil for easier installation and to avoid pinching or rolling the grommets while assembling.

Install the EGR cooler coolant vent tube.



Finishing Steps

Automotive with CM875, CM870 and CM570

- Connect the turbocharger coolant supply tube. Refer to Procedure 010-041 (</qs3/pubsys2/xml/en/procedures/35/35-010-041-tr.html>).
- Connect the EGR valve coolant supply tube to the EGR cooler coolant supply upper tube. Refer to Procedure 011-



030 (/qs3/pubsys2/xml/en/procedures/35/35-011-030-tr.html).

- Install the exhaust gas pressure sensor tube and retaining clip. Refer to Procedure 011-027 (/qs3/pubsys2/xml/en/procedures/35/35-011-027-tr.html).
- Install the air compressor air supply tube from the EGR mixer to the air compressor air supply tube at the rear of the engine and tighten the hose clamps. Refer to Procedure 012-109 (/qs3/pubsys2/xml/en/procedures/35/35-012-109-tr.html).
- Install the air intake piping. Refer to the OEM troubleshooting and repair manual.
- Fill the cooling system. Refer to Procedure 008-018 (/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html).
- Operate the engine until it reaches a temperature of 80°C [176°F] and check for coolant leaks.

Automotive with CM876

- Connect the turbocharger coolant supply and return tube. Refer to Procedure 010-041 (/qs3/pubsys2/xml/en/procedures/35/35-010-041-tr.html).
- Install the air intake piping. Refer to the OEM troubleshooting and repair manual.
- Fill the cooling system. Refer to Procedure 008-018 (/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html).
- Operate the engine until it reaches a temperature of 80°C [180°F] and check for coolant leaks.

